

- The ICD has remained in Idle Mode for a full [Idle Mode Duration](#).
- An attribute change needs to be reported to a subscriber.
- When the maximum reporting interval has been reached for an active subscription.
- A generated event needs to be reported to a subscriber.

When an ICD switches from Idle Mode to Active Mode, it MAY send all pending messages. These messages could be:

- Sending subscription reports and events.
- Sending Check-In messages.
- Sending Interaction Messages.

A node typically determines whether it is in Active or Idle mode based on whether it has any outstanding [Exchanges](#) in the Message Layer. While there are Exchanges active, a node typically will remain in Active mode. Once all Exchanges are closed and the node has no other outstanding reasons for staying active, a node SHOULD transition into Idle mode. An ICD node MAY use additional modes and logic for switching between them.

9.15.1.2. ICD Session Configurations

Due to the ICD operating modes, there may be an additional delay before the ICD receives a message. This causes the travel time of a message to be affected by the ICD's configuration and current operating mode.

When in [Idle Mode](#), the device will typically not receive messages at a faster frequency than the [Slow Polling Interval](#). As such, a client SHOULD NOT try to send the same message at a frequency higher than the [Slow Polling Interval](#) when the ICD is in Idle Mode. The ICD SHALL set its [SESSION_IDLE_INTERVAL](#) to a value greater than or equal to the [Slow Polling Interval](#).

When in [Active Mode](#), the device will typically not receive messages at a faster frequency than the [Fast Polling Interval](#). As such, a client SHOULD NOT try to send the same message at a frequency higher than the [Fast Polling Interval](#) when the ICD is in Active Mode. The ICD SHALL set its [SESSION_ACTIVE_INTERVAL](#) to a value greater than or equal to the [Fast Polling Interval](#).

The ICD SHALL set its [SESSION_ACTIVE_THRESHOLD](#) to the [Active Mode Threshold](#) stored in the [ICD Management cluster](#).

9.15.1.3. Check-In Protocol Support

This section describes the Check-In Protocol use case used by Intermittently Connected Devices (ICDs) to maintain a known relationship in case subscriptions with clients are lost. This includes how a client shares an [ICDToken](#) with the ICD, when Check-In messages are sent and how the Check-In Protocol requirements are respected.

The Check-In Protocol is a fail-safe mechanism which allows an Intermittently Connected Device (ICD) to notify a registered client that it is available for communication when all subscriptions between the client and ICD are lost. A subscription can be lost for several reasons, such as: